Palo Alto, California, at which the term 'open source' was first coined, and the Open Source Initiative was born. Almost every piece of software you can think of nowadays is open source.

So, how do open source programs get created and maintained? This was a difficult problem to solve at first. It was a huge undertaking to create and collaborate on open source projects. The majority of the software was packaged into tarballs and distributed via Web portals. The problem with this strategy was that it was impossible to keep track of who in the project was making which file changes. The code was frequently problematic and collaboration was difficult. To combat this, developers began developing version control software that could track changes and log the versions of each software release. CVS and Subversion were two of the first widely used version control systems. However, they were not without defects, and some of the most pressing issues with open source collaboration still needed to be addressed.

Finally, in 2005, Linus Torvalds produced another ground-breaking piece of software to aid with the maintenance of his earlier Linux software. Git is the name he gave to this version control system. Git was the first non-linear, distributed version control system of its kind, capable of easily and quickly managing a project with thousands of collaborators. Figure 6 depicts the fundamentals of Git's operation.

You can use Git to compile project files into what is known as a project 'repository'. This is the project's main repository. A collaborator can then 'fork' this repository, which is a fancy way of stating that he can make a local copy of it on his computer. The collaborator can then 'commit' the code to his local repository, which is a technique of updating and storing the changes made to the repository, after making any necessary revisions. The collaborator then sends a 'pull request' to the main repository's moderator. The moderator has the ability to examine the modifications made by the collaborator to the project and to 'merge' the collaborator's changes with the main repository. That is, once a pull request is sent, the project moderator can either accept or reject the changes made to the project.

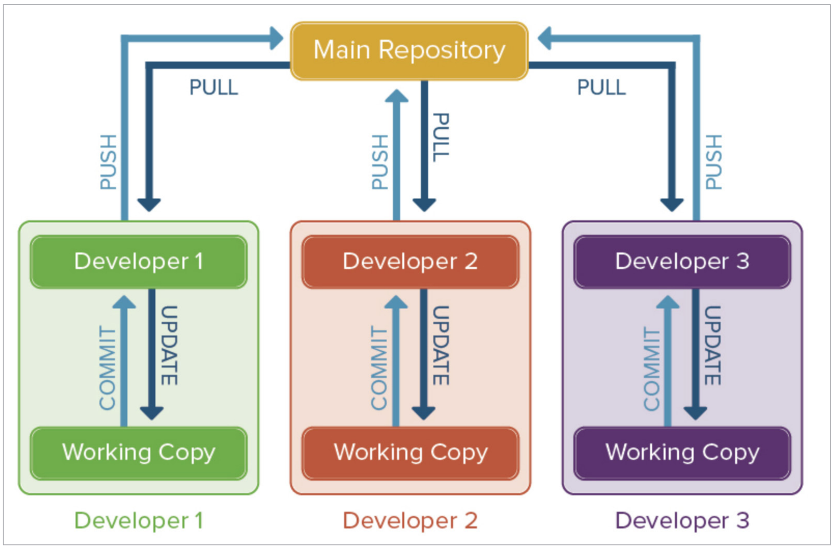

Figure 6: Working of Git (Courtesy: Google Images)

Git includes a number of other intriguing characteristics, such as 'branching', that make it a very strong and widely used open source tool. Such subtleties, however, are beyond the scope of this article.

Characteristics of blockchain

We have discussed blockchain at length in the previous sections. Let us summarise our discussion by listing the characteristics of blockchain.

Immutability: A hallmark feature of blockchain is that it is incorruptible. Data once saved in a block cannot be changed without breaking the chain.

Decentralisation: Blockchain does not require a central authority to maintain the ledger. All nodes in the network have a copy of the ledger that gets updated with every transaction.

Enhanced security: Decentralisation and consensus mechanisms built into blockchain make it highly secure, as every transaction has to be verified with 'proof of work'.

Distributed: The blockchain ledger is not a singular entity in the hands of a few; instead it is a shared entity that is accessible and maintained by everyone in the network.

Transparency: Every addition to the blockchain requires consensus from all nodes in the peer-to-peer network. This makes every transaction transparent and public.

Characteristics of open source

Similarly, let us summarise our understanding of open source by listing its characteristics.

Transparency: The founding principle of open source is to make the source code of the project public. This enhances the transparency of the project, as every detail that goes into its development can be freely accessed by anyone.

Enhanced security: Finding security loopholes in the software is way easier when the code is non-proprietary and is publicly available. This naturally enhances the security of the software or, at the very least, can help raise public awareness about the potential security related pitfalls.